

Reasoning for Code Changes

Certainly! Here are the clear, concise, and structured reasons behind the changes in the provided diff:

- **Updated `javax` to `jakarta` for Spring Boot 3 compatibility:**

- The transition from `javax` to `jakarta` is necessary for compatibility with Spring Boot 3, which uses Jakarta EE 9+.

This change affects multiple files and classes, including `User.java`, `RegisterRequestDto.java`, `Book.java`, `Tag.java`, `BorrowRecord.java`, `GlobalExceptionHandler.java`, `JwtAuthenticationEntryPoint.java`, `JwtRequestFilter.java`, and `SecurityConfig.java`.

Replaced `@Type` with `@Convert` in Hibernate 6:

- The `@Type` annotation is no longer supported in Hibernate 6. It has been replaced with `@Convert(converter = YesNoConverter.class)` to maintain the same functionality.

This change is made in `Book.java` and `BorrowRecord.java`.

Updated `WebSecurityConfigurerAdapter` and related methods for Spring Security 6:

- `WebSecurityConfigurerAdapter` is removed in Spring Security 6, so the configuration class `SecurityConfig` no longer extends it.
- The `configure(AuthenticationManagerBuilder auth)` method is replaced with a `@Bean` method for `AuthenticationManager`.
- The `configure(WebSecurity web)` method is updated to use `requestMatchers` instead of `antMatchers`.
- The `authorizeRequests` method is replaced with `authorizeHttpRequests`.

The `authenticationManagerBean` method is updated to use `authenticationConfiguration.getAuthenticationManager()`.

Fixed redundant `@Bean` annotation:

The `@Bean` annotation was mistakenly duplicated in the `authenticationManager` method. The redundant annotation was removed.

Updated `configure(WebSecurity web)` method:

The `configure(WebSecurity web)` method was defined twice. The redundant method was removed.

Fixed redundant return statements:

Redundant `return` statements in the `passwordEncoder` method were removed.

Added logging and fine handling in `BorrowService`:

- Logging statements were added to log when a book is borrowed and returned.

Fine handling was added to calculate and log fines when a book is returned overdue.

Fixed redundant `@Bean` annotation in `SecurityConfig`:

The `@Bean` annotation was mistakenly duplicated in the `passwordEncoder` method. The redundant annotation was removed.

Updated `@EnableGlobalMethodSecurity` to `@EnableMethodSecurity`:

The `@EnableGlobalMethodSecurity` annotation is deprecated in favor of `@EnableMethodSecurity` for method security in Spring Security 6.

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